

PART A

$$\begin{aligned}
 1. (a) (i) \quad \lim_{x \rightarrow 4} \frac{16-x^2}{\sqrt{x}-2} &= \lim_{x \rightarrow 4} \frac{16-x^2}{\sqrt{x}-2} \cdot \frac{\sqrt{x}+2}{\sqrt{x}+2} & (ii) \quad \lim_{x \rightarrow -\infty} \left(7 - \frac{\sqrt{5x}}{x} \right) &= \lim_{x \rightarrow -\infty} 7 - \lim_{x \rightarrow -\infty} \left(\frac{\sqrt{5x}}{x} \right) \\
 &= \lim_{x \rightarrow 4} \frac{(4-x)(4+x)(\sqrt{x}+2)}{x-4} & &= 7 - \lim_{x \rightarrow -\infty} \left(\frac{\sqrt{5x}}{\frac{x}{x}} \right) \\
 &= \lim_{x \rightarrow 4} \frac{(4-x)(4+x)(\sqrt{x}+2)}{-(4-x)} & &= 7 - \lim_{x \rightarrow -\infty} \left(\sqrt{\frac{5x}{x^2}} \right) \\
 &= \lim_{x \rightarrow 4} -(4+x)(\sqrt{x}+2) & &= 7 - \sqrt{5} \lim_{x \rightarrow -\infty} \left(\sqrt{\frac{1}{x}} \right) \\
 &= -(4+4)(\sqrt{4}+2) & &= 7 - \sqrt{5}(0) \\
 &= -32 & &= 7
 \end{aligned}$$

$$\begin{aligned}
 (b) \quad \lim_{x \rightarrow 1} \frac{f(x)-2}{3-x} &= 5 \\
 \frac{\lim_{x \rightarrow 1} [f(x)-2]}{\lim_{x \rightarrow 1} [3-x]} &= 5 \\
 \lim_{x \rightarrow 1} [f(x)-2] &= 5 \lim_{x \rightarrow 1} [3-x] \\
 \lim_{x \rightarrow 1} f(x) - \lim_{x \rightarrow 1} 2 &= 5 \lim_{x \rightarrow 1} [3-x] \\
 \lim_{x \rightarrow 1} f(x) &= 2 + 5(3-1) \\
 \lim_{x \rightarrow 1} f(x) &= 12
 \end{aligned}$$

$$\begin{aligned}
 2.(a) \quad f(x) &= \frac{1}{\sqrt{2-x}} \quad , \quad f(x+h) = \frac{1}{\sqrt{2-(x+h)}} \\
 f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{\frac{1}{\sqrt{2-(x+h)}} - \frac{1}{\sqrt{2-x}}}{h} \\
 &= \lim_{h \rightarrow 0} \frac{1}{h} \left[\frac{\sqrt{2-x} - \sqrt{2-(x+h)}}{(\sqrt{2-x})(\sqrt{2-(x+h)})} \cdot \frac{\sqrt{2-x} + \sqrt{2-(x+h)}}{\sqrt{2-x} + \sqrt{2-(x+h)}} \right] \\
 &= \lim_{h \rightarrow 0} \frac{1}{h} \left[\frac{2-x - [2-(x+h)]}{(\sqrt{2-x})(\sqrt{2-(x+h)})(\sqrt{2-x} + \sqrt{2-(x+h)})} \right] \\
 &= \lim_{h \rightarrow 0} \frac{1}{h} \left[\frac{h}{(\sqrt{2-x})(\sqrt{2-(x+h)})(\sqrt{2-x} + \sqrt{2-(x+h)})} \right] \\
 &= \lim_{h \rightarrow 0} \frac{1}{(\sqrt{2-x})(\sqrt{2-(x+h)})(\sqrt{2-x} + \sqrt{2-(x+h)})} \\
 &= \frac{1}{\sqrt{(2-x)^2} (2\sqrt{2-x})} = \frac{1}{2(2-x)^{\frac{3}{2}}}
 \end{aligned}$$

$$\begin{aligned}
 \text{(b) (i)} \quad y &= \ln \sqrt{(4x-5)^3} = \ln (4x-5)^{\frac{3}{2}} \\
 &= \frac{1}{(4x-5)^{\frac{3}{2}}} \frac{d}{dx} (4x-5)^{\frac{3}{2}} \\
 &= \frac{1}{(4x-5)^{\frac{3}{2}}} \left[\frac{3}{2} (4x-5)^{\frac{1}{2}} (4) \right] \\
 &= \frac{6}{4x-5}
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii)} \quad y &= \cot^3(7x^2 + 4x) \\
 &= 3\cot^2(7x^2 + 4x) \left[-\operatorname{cosec}^2(7x^2 + 4x) \right] [14x + 4] \\
 &= -6(7x + 2) \cot^2(7x^2 + 4x) \operatorname{cosec}^2(7x^2 + 4x)
 \end{aligned}$$

$$\begin{aligned}
 3. \quad f(x) &= \frac{2}{x^2 + 5} \\
 f(x) &= 2(x^2 + 5)^{-1} \\
 f'(x) &= -(2x^2 + 5)^{-2} (2x) \\
 &= -\frac{4x}{(x^2 + 5)^2}
 \end{aligned}$$

$$f'(x) = 0$$

$$-\frac{4x}{(x^2 + 5)^2} = 0$$

$$x = 0$$

$$x = 0, f(0) = \frac{2}{5} \quad \therefore \left(0, \frac{2}{5} \right)$$

	$x < 0$	$x > 0$
$\frac{dy}{dx}$	+	-
Sketch of tangent		

$\therefore \left(0, \frac{2}{5} \right)$ is a relative maximum point

PART B

$$\begin{aligned}
 1. (i) \quad z &= \frac{(2-i)^2}{1+i} = \frac{4-4i+i^2}{1+i} \\
 &= \frac{3-4i}{1+i} \times \frac{1-i}{1-i} \\
 &= \frac{3-7i+4i^2}{1+1} \\
 &= \frac{-1-7i}{2} = -\frac{1}{2} - \frac{7i}{2}
 \end{aligned}$$

$$\begin{aligned}
 (ii) \quad |z| &= \sqrt{\left(-\frac{1}{2}\right)^2 + \left(-\frac{7}{2}\right)^2} \\
 &= \sqrt{\frac{1+49}{4}} \\
 &= \sqrt{\frac{50}{4}} \\
 &= \frac{5\sqrt{2}}{2}
 \end{aligned}$$

$$\begin{aligned}
 \alpha &= \tan^{-1}\left(\frac{7/2}{1/2}\right) \\
 &= \tan^{-1}(7) \\
 &= 1.4289 \text{ rad} \\
 \therefore \arg(z) &= -\pi + 1.4289 \text{ rad} \\
 &= -1.7127 \text{ rad}
 \end{aligned}$$

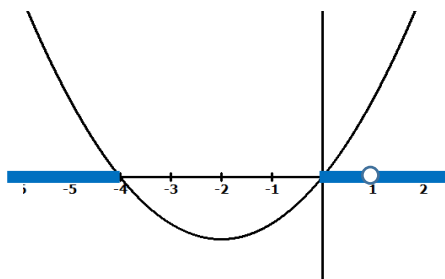
$$(iii) \quad z = \frac{5\sqrt{2}}{2} (\cos(-1.7127 \text{ rad}) + i \sin(-1.7127 \text{ rad}))$$

$$\begin{aligned}
 2. (a) \quad \frac{8-2(x+4)}{x+4} &\leq 0 \\
 \frac{-2x}{x+4} &\leq 0 \\
 \frac{2x}{x+4} &\geq 0
 \end{aligned}$$

	$-\infty$	-4	0	∞
$2x$	-	-	+	
$x+4$	-	+	+	
$\frac{2x}{x+4}$	+	-	+	

$\therefore$ The solution is $(-\infty, -4) \cup [0, \infty)$

$$\begin{aligned}
 (b) \quad |3x+2| &\geq 2|x-1|, x \neq 1 \\
 (3x+2)^2 &\geq 4(x-1)^2 \\
 9x^2 + 12x + 4 &\geq 4x^2 - 8x + 4 \\
 5x^2 + 20x &\geq 0 \\
 5x(x+4) &\geq 0
 \end{aligned}$$



$\therefore$ The solution is $(-\infty, -4] \cup [0, 1) \cup (1, \infty)$

(b) Alternative method

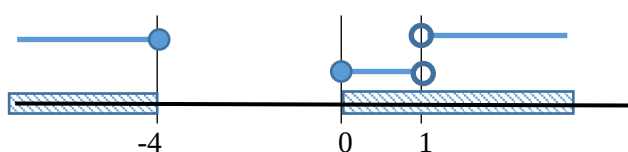
$$\begin{aligned} \frac{3x+2}{x-1} &\geq 2 & \text{or} & & \frac{3x+2}{x-1} &\leq -2 \\ \frac{3x+2-2(x-1)}{x-1} &\geq 0 & & & \frac{3x+2+2(x-1)}{x-1} &\leq 0 \\ \frac{x+4}{x-1} &\geq 0 & & & \frac{5x}{x-1} &\leq 0 \end{aligned}$$

$x+4$	-	+	+
$x-1$	-	-	+
$\frac{x+4}{x-1}$	+	-	+

$$(-\infty, -4] \cup (1, \infty)$$

$5x$	-	+	+
$x-1$	-	-	+
$\frac{5x}{x-1}$	+	-	+

$$[0, 1)$$



$\therefore$ The solution is $(-\infty, -4] \cup [0, 1) \cup (1, \infty)$

3. (i) $f(x) = x^2 - 2x - 3, x \leq 1$

$$= (x-1)^2 - 4$$

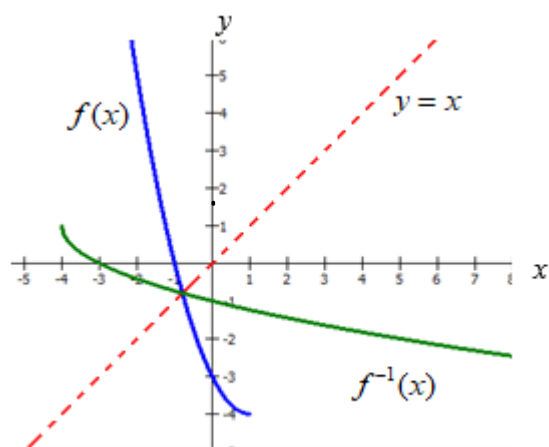
$$ff^{-1}(x) = x$$

$$(f^{-1}(x) - 1)^2 - 4 = x$$

$$f^{-1}(x) - 1 = -\sqrt{x+4} \quad (\text{because the } D_f \text{ is } (-\infty, 1])$$

$$\therefore f^{-1}(x) = 1 - \sqrt{x+4}$$

(ii)



(iii) $D_{f^{-1}} = [-4, \infty)$, $R_{f^{-1}} = (-\infty, 1]$

$$4. (i) \quad f(x) = \ln(3x+2)$$

$$ff^{-1}(x) = x$$

$$\ln(3f^{-1}(x)+2) = x$$

$$3f^{-1}(x)+2 = e^x$$

$$\therefore f^{-1}(x) = \frac{e^x - 2}{3}$$

$$(ii) \quad g(x) = e^x + 2$$

$$gg^{-1}(x) = x$$

$$e^{g^{-1}(x)} + 2 = x$$

$$e^{g^{-1}(x)} = x - 2$$

$$g^{-1}(x) = \ln(x-2)$$

$$(iii) \quad f^{-1}(x) = \frac{e^x - 2}{3}, \quad g^{-1}(x) = \ln(x-2)$$

$$\begin{aligned} (g \circ f)^{-1}(x) &= f^{-1}g^{-1}(x) \\ &= f^{-1}(\ln(x-2)) \\ &= \frac{e^{\ln(x-2)} - 2}{3} \\ &= \frac{x-4}{3} \end{aligned}$$

$$5. (a)$$

$$f(x) = \begin{cases} \frac{x^2 + 3x - 4}{|x-1|}, & x < 1 \\ 5, & x = 1 \\ 2x - 7, & x > 1 \end{cases} \quad |x-1| = \begin{cases} x-1, & x > 1 \\ -(x-1), & x < 1 \end{cases}$$

$$\begin{aligned} \lim_{x \rightarrow 1^-} \frac{x^2 + 3x - 4}{-(x-1)} &= \lim_{x \rightarrow 1^-} \frac{(x+4)(x-1)}{-(x-1)} \\ &= \lim_{x \rightarrow 1^-} -(x+4) \\ &= -(1+4) \\ &= -5 \end{aligned}$$

$$\begin{aligned} \lim_{x \rightarrow 1^+} 2x - 7 &= 2(1) - 7 \\ &= -5 \end{aligned}$$

$$\begin{aligned} \lim_{x \rightarrow 1^+} f(x) &= \lim_{x \rightarrow 1^+} f(x) = -5 \\ \therefore \lim_{x \rightarrow 1} f(x) &= -5 \\ f(1) &= 5 \end{aligned}$$

$$\text{Since } \lim_{x \rightarrow 1} f(x) \neq f(1) \quad \therefore f(x) \text{ is not continuous at } x=1$$

$$(b) \quad g(x) = \begin{cases} e^x + 2p & , \quad x < 0 \\ 3x^2 + 2x - 1 & , \quad 0 \leq x < 3 \\ -8q & , \quad x \geq 3 \end{cases}$$

$$\lim_{x \rightarrow 0^-} g(x) = \lim_{x \rightarrow 0^+} g(x)$$

$$\lim_{x \rightarrow 0^-} e^x + 2p = \lim_{x \rightarrow 0^+} 3x^2 + 2x - 1$$

$$e^0 + 2p = -1$$

$$2p = -2$$

$$p = -1$$

$$\lim_{x \rightarrow 3^-} g(x) = \lim_{x \rightarrow 3^+} g(x)$$

$$\lim_{x \rightarrow 3^-} 3x^2 + 2x - 1 = \lim_{x \rightarrow 3^+} (-8q)$$

$$3(3)^2 + 2(3) - 1 = -8q$$

$$-8q = 32$$

$$q = -4$$

$$6. (a) \quad \ln y = x^2 e^{3x} + 4y$$

$$u = x^2$$

$$v = e^{3x}$$

$$\frac{1}{y} \frac{dy}{dx} = 3x^2 e^{3x} + 2xe^{3x} + 4 \frac{dy}{dx}$$

$$u' = 2x$$

$$v' = 3e^{3x}$$

$$\left[\frac{1}{y} - 4 \right] \frac{dy}{dx} = 3x^2 e^{3x} + 2xe^{3x}$$

$$\left[\frac{1-4y}{y} \right] \frac{dy}{dx} = 3x^2 e^{3x} + 2xe^{3x}$$

$$\frac{dy}{dx} = \frac{y(3x^2 e^{3x} + 2xe^{3x})}{1-4y}$$

$$\frac{dy}{dx} = \frac{xye^{3x}(3x+2)}{1-4y}$$

$$(b) (i) \quad x = 2t^3 \quad y = t^2 - 4t$$

$$\frac{dx}{dt} = 6t^2 \quad \frac{dy}{dt} = 2t - 4$$

$$\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx}$$

$$= (2t - 4) \left(\frac{1}{6t^2} \right)$$

$$= \frac{2(t-2)}{6t^2}$$

$$= \frac{t-2}{3t^2}$$

$$\begin{aligned}
 \frac{d^2 y}{dx^2} &= \frac{\frac{d}{dt} \left[\frac{dy}{dx} \right]}{\frac{dx}{dt}} = \frac{\frac{d}{dt} \left(\frac{t-2}{3t^2} \right)}{\frac{dx}{dt}} \\
 &= \frac{3t^2 - 6t(t-2)}{(3t^2)^2} \times \frac{1}{6t^2} & \begin{array}{ll} u = t-2 & v = 3t^2 \\ u' = 1 & v' = 6t \end{array} \\
 &= \frac{3t(4-t)}{54t^6} \\
 &= \frac{4-t}{18t^5}
 \end{aligned}$$

$$(ii) \quad \frac{dy}{dx} = 0$$

$$\frac{t-2}{3t^2} = 0$$

$$t = 2, \quad x = 2(2)^3 = 16$$

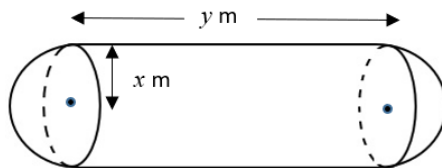
$$y = 2^2 - 4(2) = -4$$

$$\therefore \text{Point } (16, -4)$$

When $x \rightarrow \infty$, $t \rightarrow \infty$

$$\begin{aligned}
 \lim_{x \rightarrow \infty} \frac{d^2 y}{dx^2} &= \lim_{t \rightarrow \infty} \frac{4-t}{18t^5} = \lim_{t \rightarrow \infty} \frac{\frac{4}{t^5} - \frac{t}{t^5}}{\frac{18t^5}{t^5}} \\
 &= \lim_{t \rightarrow \infty} \frac{\frac{4}{t^5} - \frac{1}{t^4}}{18} \\
 &= \frac{0-0}{18} \\
 &= 0
 \end{aligned}$$

7.



- (a) Tot. surface area, $A = \text{Area of 2 hemispheres} + \text{Area of the cylinder}$

$$= 2(2\pi x^2) + 2\pi xy$$

$$= 4\pi x^2 + 2\pi xy$$

$$(b) \quad \text{Volume of the whole solid} = \frac{\pi}{3} \text{m}^3$$

$$2\left(\frac{2}{3}\pi x^3\right) + \pi x^2 y = \frac{\pi}{3}$$

$$\frac{4}{3}\pi x^3 + \pi x^2 y = \frac{\pi}{3}$$

$$4\pi x^3 + 3\pi x^2 y = \pi$$

$$3\pi x^2 y = \pi - 4\pi x^3$$

$$y = \frac{\pi(1-4x^3)}{3\pi x^2}$$

$$y = \frac{1-4x^3}{3x^2}$$

$$(c) \quad A = 4\pi x^2 + 2\pi xy$$

$$= 4\pi x^2 + 2\pi x \left(\frac{1-4x^3}{3x^2} \right)$$

$$= 4\pi x^2 + \frac{2\pi x}{3x^2} - \frac{8\pi x^4}{3x^2}$$

$$= 4\pi x^2 + \frac{2\pi}{3x} - \frac{8\pi x^2}{3}$$

$$= \frac{2\pi x}{3x^2} + \frac{4\pi x^4}{3x^2}$$

$$A = \frac{2\pi}{3x} + \frac{4\pi x^2}{3} \quad (\text{Shown})$$

$$A = \frac{2\pi}{3x} + \frac{4\pi x^2}{3}$$

$$\frac{dA}{dx} = -\frac{2\pi x^{-2}}{3} + \frac{4}{3}(2)\pi x$$

$$= -\frac{2\pi}{3x^2} + \frac{8}{3}\pi x$$

$$\frac{dA}{dx} = 0, \quad -\frac{2\pi}{3x^2} + \frac{8}{3}\pi x = 0$$

$$\frac{2\pi}{3x^2} = \frac{8}{3}\pi x$$

$$6\pi = 24\pi x^3$$

$$x^3 = \frac{1}{4}$$

$$x = 0.63$$

$$\frac{d^2 A}{dx^2} = \frac{4\pi}{3x^3} + \frac{8}{3}\pi$$

$$x = 0.63, \quad \frac{d^2 A}{dx^2} = \frac{4\pi}{3(0.63)^3} + \frac{8}{3}\pi$$

$$= 25.13 (> 0) \quad \text{min}$$

Total surface area, A is minimum when $x = 0.63$

$$\text{Area minimum} = \frac{2\pi}{3(0.63)} + \frac{4\pi(0.63)^2}{3}$$

$$= 4.987 \text{ m}^2$$